

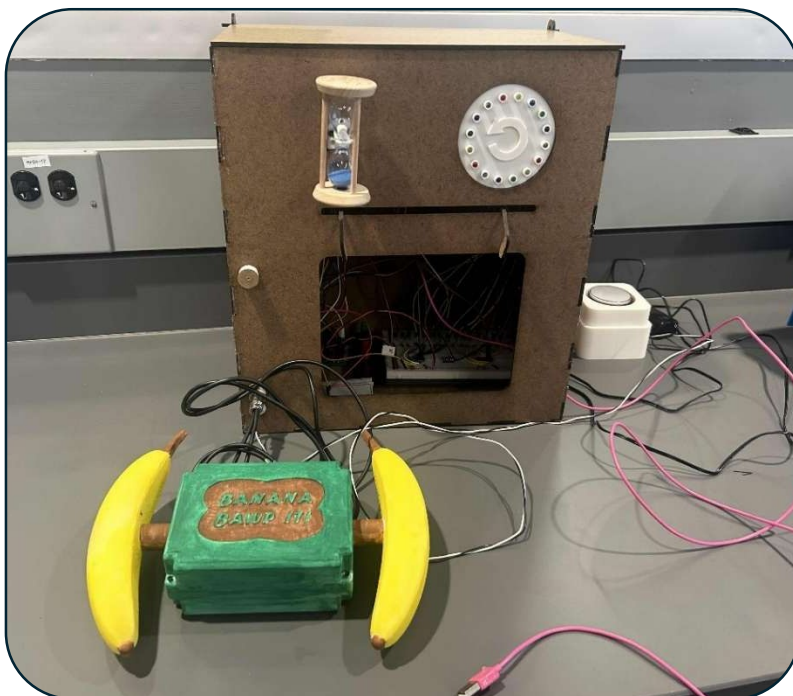
Banana Bawp It!

What?

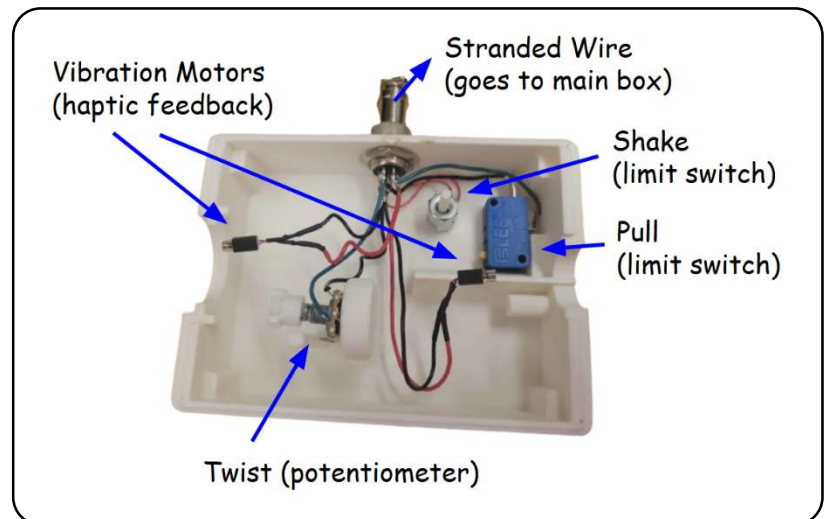
- The **Banana Bawp It!** is a jungle themed “Bop-It” game that prompts the user to fulfill certain actions to gain points

Challenge

- Design an **automated, walk-up interactive device** that demonstrates technology as playful, silly, and engaging
- Incorporate distinct user interactions using **diverse sensing modalities**—including **analog** and **non-contact sensors**—with **real-time feedback**
- Implement a **time-driven interaction sequence** with an **idle welcoming mode** and **automatic reset** on user inactivity



Assembly of Product



Results

- Designed a **potentiometer-based “twist”** interaction **coupled to a circular LED array** that responds proportionally to user rotation of banana handles
- Implemented a **“pull” input** via banana handles using **limit switches**
- Designed a custom **“shake” interaction** using loose nuts to **generate multiple rising-edges**
- Developed **event-driven** embedded software using a **state machine** to trigger immediate **audio cues** for each user action

DESIGN

- Controller designed in **SolidWorks**
- Main enclosure **laser cut**
- Low-level embedded C** firmware
- Aviation connector** used for robust controller wiring